Vaskar Chakma

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Education

- **Chung-Ang University, College of Software, Dept. of Computer Science and Engineering** — Seoul, Republic of Korea
  Doctor of Philosophy (Ph.D.) Degree in Computer Science and Engineering (Big Data) — 2025.09.01 - Present
  Advisor: Professor Wooyeol Choi
- **Nantong University, School of Artificial Intelligence and Computer Science** — Nantong, Jiangsu, China
  Bachelor of Engineering Degree in Computer Science and Technology — 2021.09.10 - 2025.06.19
  GPA: 3.86/5.00, WES converted: 3.37/4.0, Ranking: 2/32
  Thesis: CardioForest: An Explainable Ensemble Learning Model for Automatic Wide QRS Complex Tachycardia Diagnosis from ECG
  Advisors: Professor Gao Zhan, Professor Xiaolin Ju

Skills Summary

- **Programming**: Python, C, MATLAB, [Web Dev: HTML/CSS/JS]
- **AI/ML**: TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, XAI [SHAP]
- **Tools**: Overleaf (Latex), MATLAB, Origin, Visual Studio Code, Google Colab, Git, Microsoft Office, Unity (Vuforia Engine), Nmap
- **Languages**: Chakma (Native), Bengali (Native), English (Advanced), Chinese (Intermediate), Hindi (Intermediate), Urdu (Intermediate), Korean (Fundamental)

Experience

- **Chung-Ang University** — Seoul, Republic of Korea
  College of Software, School of Computer Science and Engineering
  - **Graduate Research Assistant (GRA)**: Intelligent Networking Lab (INL) — 2025.09 - Present
  - **Laboratory Assistant (LA)**: Coding Bootcamp Lab [B208, R519] — 2026.03 - 2027.02
  - **Teaching Assistant (TA)**: (1) [Fall 25, 26] Programming Course (UG) [47710] — 2025.09 - 2026.12; (2) [Spring 26] Network Applications and Design (UG) [34579] — 2026.03 - 2026.08
  - **Mentorship**: (1) Winter Undergraduate Research Program — 2026.01 - 2026.02; (2) Summer Undergraduate Research Program — 2026.07 - 2026.08
- **Nantong University** — People's Republic of China
  Student Counselor, School of International Education — 2023.09.02 - 2025.06.19

Academic Services

- **Subreviewer / External Reviewer** — Duy Tan University, Da Nang, Vietnam
  The 11th International Conf. on Next Generation Computing (ICNGC 2025) — 2025.12.17 - 2025.12.20

Research Projects

- **[PR] Research on Full-Stack Networking Technologies for 6G AI-Native NTN** — Seoul, Republic of Korea
  Funded by National Research Foundation of Korea (NRF) — 2026.03 - 2029.02
- **[PR] BK21 FOUR (CAU Institute for Innovative Talent of Big Data)** — Seoul, Republic of Korea
  Funded by National Research Foundation of Korea (NRF) — 2025.09 - 2027.08

Journal Publications

- **[JIMH'26]**: **Vaskar Chakma**, Xiaolin Ju, Heling Cao, Xue Feng, Ji Xiaodong, Pan Haiyan, and Gao Zhan. 'CardioForest: An Explainable Ensemble Learning Model for Automatic Wide QRS Complex Tachycardia Diagnosis from ECG'. Journal of Intelligent Medicine and Healthcare, 4(1), 37–86. [DOI: 10.32604/jimh.2026.075201]
- **[JIMH'26]**: **Vaskar Chakma**, MD Jaheid Hasan Nerab, Abdur Rouf, Abu Sayed, Hossem MD Saim, and Md Nournabi Khan. 'Machine Learning Models for Predicting Smoking-Related Health Decline and Disease Risk'. Journal of Intelligent Medicine and Healthcare, 4(1), 1–35. [DOI: 10.32604/jimh.2026.074347]
- **[IST'25]**: Ju Xiaolin, Yi Cao, Xiang Chen, Lina Gong, **Vaskar Chakma**, and Xin Zhou. 'JIT-CF: Integrating Contrastive Learning with Feature Fusion for Enhanced Just-in-Time Defect Prediction'. Information and Software Technology, 182 (2025): 107706. [DOI: 10.1016/j.infsof.2025.107706]
- **[MRE'25]**: Zhou ZhiGuang, Jagaran Chakma, Md Ahatasamul Hoque, **Vaskar Chakma**, and Asif Ahmed. 'Prediction of UHPC Mechanical Properties Using Optimized Hybrid Machine Learning Model with Robust Sensitivity and Uncertainty Analysis'. Materials Research Express 12, no. 8 (2025): 085703. [DOI: 10.1088/2053-1591/adf8c4]
- **[JAIDM'25]**: Xiaolin Ju, **Vaskar Chakma**, Misbahul Amin, and Arkhid Chakma Joy. 'What Drives House Prices? A Linear Regression Approach to Size, Condition, and Features'. Journal of Artificial Intelligence & Data Mining, 13(1), 41-51, 2025. [DOI: 10.22044/jadm.2025.15529.2668]
- **[EJTAS'25]**: **Vaskar Chakma**, Amin Misbahul, Rouf Abdur, Suruj Al Mahmud, Mia Raju, and Rafid Mustavi. 'From Margins to Mainstream: Can Artificial Intelligence (AI) Reshape Governance for Chittagong Hill Tracts Indigenous Communities?'. EJTAS, 3(1), 166-178, 2025. [DOI: 10.59324/ejtas.2025.3(1).16]

Conference Publications

- **International Conferences**:
  - **[ICOIN'27]**: **Vaskar Chakma**, and Wooyeol Choi. 'Which Satellite to Refresh Next? Look-Ahead Scheduling of Orbit Data for Timing Advance in LEO Networks'. The 41st International Conference on Information Networking, Nha Trang, Vietnam [Jan 13 - 15, 2027]
  - **[ICNGC'26]**: **Vaskar Chakma**, and Wooyeol Choi. 'Graph-Based Attribution of Multi-Tenant Resource Contention in AI-RAN Infrastructure'. The 12th International Conference on Next Generation Computing, Hanoi, Vietnam [Nov 25 - 28, 2026]
  - **[ICUFN'26]**: **Vaskar Chakma**, and Wooyeol Choi. 'Self-Directed Spectrum Allocation Framework for Integrated TN-NTN 6G Networks'. The 17th International Conference on Ubiquitous and Future Networks, Milan, Italy [July 7 - 10, 2026] [DOI: 10.1109/ICUFN69619.2026.11628659]
  - **[ICAIIC'26]**: **Vaskar Chakma**, Seong-Yong Moon, and Wooyeol Choi. 'Deep Learning-Based Traffic Prediction for Non-Terrestrial Networks: A Hybrid Satellite-UAV Approach'. The 8th International Conference on Artificial Intelligence in Information and Communication, Tokyo University of Science, Tokyo, Japan [February 24 - 27, 2026] [DOI: 10.1109/ICAIIC68212.2026.11454384]
  - **[ICTC'25]**: **Vaskar Chakma**, and Wooyeol Choi. '6G Enabled Digital Twin Framework for Real-Time Cyber-Physical Systems'. ICTC Workshop on Big Data, CPS, and 5G & 6G Communication Networks (IWBCN), Jeju, Republic of Korea [October 14 - 17, 2025]
- **Domestic Conference**:
  - **[JCCI'26]**: **Vaskar Chakma**, and Wooyeol Choi. 'RAN-Aware Adversarial Training Closes the Robustness Gap in Near-RT RIC Handover Control', The 36th Joint Conference on Communications and Information, The Ocean Resort, Yeosu, Republic of Korea [April 15 - 17, 2026] [DOI: 10.5281/zenodo.18870997]

Invited Talk

- [CAU Talk'26] Graphic Deep Reinforcement Learning for Dynamic Resource Allocation in Space-Air-Ground Integrated Networks

Awards

- [Jun. 2026] President's *Excellence Award and the People's Choice Award* - Chung-Ang University 3MT Competition
- [Jul. 2024] Outstanding Student, '*Space, Sky, Land, and Sea Integrated Network Summer School*' - Nantong University
- [Nov. 2023] Top 3 in *Chinese Proficiency Test* – Nantong University

Scholarships

- **[CAU] Chung-Ang University**:
  [1] Chung-Ang University Young Scientist Scholarship (CAYSS)
  [2] Industry-Academic Cooperation Foundation Scholarship
  [3] Brain Korea-21 Graduate School Innovation Scholarship
- [~~HUST 2025~~] *Chinese Government Scholarship*, accepted as a Master's Student (CST), Huazhong University of Science and Technology (HUST), People's Republic of China
- [~~Fudan 2025~~] *Shanghai Government Scholarship*, accepted as a Master's Student (Artificial Intelligence), Fudan University, People's Republic of China
- [NTU] *Nantong University Bachelor Scholarship*, Jiangsu Provincial Government, People's Republic of China

Research Interests

AI-RAN (Radio Access Network), TN-NTN, Wireless Networking, Interdisciplinary Artificial Intelligence, Quantum Machine Learning (QML)